INTERFACE CONTROL DOCUMENT

SEPTA

FRICTION AND PROPULSION LONWORKS INTERFACE

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REVISIONS

REV.	DESCRIPTION

A06986530000-0506.doc

Prepared by: DES 01/16/07 Approved by: RJM 01/18/07

A Section 2 Revised last sentence to remove 'assumed….text' as this is a dedicated network between the WPT and MELCO systems.

Section 3 Updated referenced information for HCN-L5772-24 Brake Interface between Propulsion Converter/Inverter and Friction Brake Control Unit.

Section 5.2.1 Added Error Handling for Data value and message. Updated Values text for AW0 and AW3 range.

Section 5.3.1 Added Error Handling for Data value and message. Updated Values text for jerk limiting.

Section 5.3.2.3 Changed minDelta preset value to SNVT value change is 66 (0.3324%).

Section 5.4.1 Inserted Bit '1' as Requesting Dynamic Effort field. Updated the remaining bits for this status bit table.

Section 5.5.1 Added Error Handling for Data value and message. Updated Values text for jerk limiting.

Section 5.6.1 Revised Bit '0' as WSP Activity A End. Inserted Bit '1' for WSP Activity F End. Inserted Bit '4' as Providing Dynamic Effort field. Updated the remaining bits for this status bit table. All other bit fields renumbered.

Section 5.8.1 Added Error Handling for Data value and message. Updated Values text for AW0 and AW3 range.

Section 5.9.1 Added Error Handling for Data value and message. Updated Values text for jerk limiting.

Section 5.10.1 Inserted Bit '1' as Requesting Dynamic Effort field. Updated the remaining bits for this status bit table.

Section 5.11.1 Added Error Handling for Data value and message. Updated Values text for jerk limiting.

Section 5.11.2.3 Changed minDelta preset value to SNVT value change is 66 (0.3324%)

Section 5.12.1 Revised Bit '0' as WSP Activity A End. Inserted Bit '1' for WSP Activity F End. Inserted Bit '4' as Providing Dynamic Effort field. Updated the remaining bits for this

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status bit table. All other bit fields renumbered.

Prepared by: DES 05/08/07 Approved by: RBZ 05/09/07

B Revision below are based on SEPTA comments in SC00327.

Revised section 2.0 to include references to standards IEEE 1473, ANSI/EIA/CEA 709, UTS REDE101463.

Revised standards list and replaced MELCO reference document with UTS document in section 3.0.

Updated Table 1 to assign node addresses.

Updated section 4.3.5 to include action on loss of message acknowledgement.

Updated 4.3.6.2 to clarify Maximum Send Time purpose.

Updated 4.3.8 to clarify direct network wiring interface.

Added sections 4.3.13 through 4.3.15.

Revised title of section 5.0, inserted section 6.0 prior to old section 5.7, renumbering all subsequent sections.

Revised section 5.2.1 to clarify error handling requirement.

Revised 5.3.1 to specify propulsion system performance of jerk limiting.

Revised 5.5.1 to specify propulsion system performance of jerk limiting and clarification of error handling.

Revised section 6.3.1 to specify propulsion system performance of jerk limiting.

Revised Table 13 and added Table 14 to specify network variable configuration.

Prepared by: DES 11/01/07 Approved by: RBZ 11/02/07

C Section 4.3.3 Updated Table 3 FBCU node priority value to be 2 instead of 0. Table 4 C/I node priority value to be 3 instead of 0. 0 and 1 are reserved, FBCU and CI must be different settings.

Section 5 Revised 10 msec minimum send time value to 20 msec which is consistent with the FBCU and C/I program logic. Added text for the handling requirements for new and old data.

Section 5.2.1 Revised ‘Values’ paragraph for clarification. Removed statement of the expected AW0 and AW3 values as 10% and 74%. Range is defined as 10% to 90%

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corresponding to 62.142 metric tonnes to 85 metric tonnes. Added standard English lbs values.

Section 5.2.2.1 Added 'current' to the text in the 1st sentence of the description for transmitted data. Revised network configuration setting value to be limited to 65000 sec consistent with the program logic constraints.

Section 5.2.2.2 Revised min send time to 20 msec.

Section 5.2.2.3 Clarified preset value to 100 counts representing 0.5%
Description – clarified paragraph and removed last sentence regarding the min delta value.

Section 5.3.1 Updated to include standard English force lbs values.

Section 5.3.2.1 Added 'current' to the text in the 1st sentence of the description for transmitted data. Revised network configuration setting value to be limited to 65000 sec consistent with the program logic constraints.

Section 5.3.2.2 Revised min send time to 20 msec.

Section 5.3.2.3 Clarified preset value to 60 counts representing 0.3%
Description – clarified paragraph and removed last sentence regarding the min delta value.

Section 5.4.1 Added bit '2' for indication of brake mode and coast mode. Clarified bit '1' information.

Section 5.4.2.1 Added 'current' to the text in the 1st sentence of the description for transmitted data. Revised network configuration setting value to be limited to 65000 sec consistent with the program logic constraints.

Section 5.4.2.2 Revised min send time to 20 msec.

Section 5.5.1 revised text for out of range in the Values section and Error handling section. Replaced 'send' with 'receive' in the max time text for Error handling. Updated to include standard English force lbs values.

Section 5.5.2.2 Revised max receive time to 500 msec to be consistent with 2x to 3x max send time values. Revised network configuration setting value to be limited to 65000 sec consistent with the program logic constraints.

Section 5.6.1 Clarified bit '4' information.

Section 5.6.2.2 Revised max receive time to 500 msec to be consistent with 2x to 3x max send time values. Revised network configuration setting value to be limited to 65000 sec consistent with the program logic constraints.

Section 6.0 Added information regarding propulsion objects and design requirements for handling of new data.

Section 6.2.1 Revised 'Values' paragraph for clarification. Removed statement of the

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expected AW0 and AW3 values as 10% and 74%. Range is defined as 10% to 90% corresponding to 62.142 metric tonnes to 85 metric tonnes. Added standard English lbs values.

Section 6.2.2.2 Revised max receive time to 500 msec to be consistent with approximately 2x to 3x max send time values. Revised network configuration setting value to be limited to 65000 sec consistent with the program logic constraints.

Section 6.3.1 Revised Values and Error Handling section for out of range handling to use 0 lbs [0 kN] and based on max receive time rather than max send time. Updated to include standard English force lbs values.

Section 6.3.2.2 Revised max receive time to 500 msec to be consistent with approximately 2x to 3x max send time values. Revised network configuration setting value to be limited to 65000 sec consistent with the program logic constraints.

Section 6.4.1 Added bit '2' for indication of brake mode and coast mode. Clarified bit '1' information.

Section 6.4.2.2 Revised max receive time to 500 msec to be consistent with approximately 2x to 3x max send time values. Revised network configuration setting value to be limited to 65000 sec consistent with the program logic constraints.

Section 6.5.1 Revised Values and Error Handling section for out of range handling to use 0 lbs 0 kN and based on max receive time rather than max send time. Updated to include standard English force lbs values.

Section 6.5.2.1 Revised max send time to 200 msec to be consistent with max receive times as approximately 2x to 3x values. Revised network configuration setting value to be limited to 65000 sec consistent with the program logic constraints.

Section 6.5.2.2 Revised min send time to 20 msec.

Section 6.5.2.3 Clarified preset value to 60 counts representing 0.3%

Section 6.6.1 Clarified bit '4' information.

Section 6.6.2.1 Revised max send time to 200 msec to be consistent with max receive times as approximately 2x to 3x values. Revised network configuration setting value to be limited to 65000 sec consistent with the program logic constraints.

Section 6.6.2.2 Revised min send time to 20 msec.

Section 7 Updated Table 13 for Max, Min period times to agree with new settings. Updated Table 14 to agree with C/I Index Address programmed settings. Updated Max, Min period times to agree with new settings.

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1.0 PURPOSE

The purpose of this document is to provide the interface function for the LonWorks® network connection on the SEPTA project for the interface between the Friction brake system and the Propulsion brake system.

2.0 SCOPE

This document provides information proposed for a LonWorks interface between the Wabtec Passenger Transit (WPT) Friction Brake Control Unit (FBCU) and the MELCO Propulsion equipment for the purpose of brake blending on a car level. The signal interfaces described in this document are based on those signals as defined by UTS in REDE101463 Brake Interface between CI and FBEU.

The network interface between Propulsion and Friction Brake is a dedicated LonWorks network that is not shared with any other system. All information between the two subsystems will use the LonWorks network, and there are no other electrical signals between the two subsystems.

The network shall implement the IEEE Std 1473-L (EIA/CEA-709.1) network standards as described in this document.

3.0 RELATED DOCUMENTS

LonTalk Protocol LonWorks Engineering Bulletin EB 161

Echelon Design Practices for Using a LonWorks Free Topology Channel on Trains

LonWorks FTT-10A Free Topology Transceiver User's Guide, v5, 078-0156-01

SNVT Master List and Programmer's Guide, 005-0027-01

REDE101463 Brake Interface between CI and FBEU, United Transit Systems

IEEE Std 1473-1999 IEEE Standard for Communications Protocol Aboard Trains

ANSI/EIA/CEA-709.1 Revision: B Date: 09/13/02 Control Network Protocol Specification

ANSI/EIA-709.2 Free Topology Twisted Pair Channel Specification.

4.0 NODE ATTRIBUTES AND SETTINGS

The FBCU and MELCO Propulsion Equipment is a multi object node which provides control and status information.

4.1 Node Description

The equipment has an electronic module that makes use of a LonWorks neuron chip to provide its function and the communication interface. This node is set for static installation.

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4.2 Node Documentation

This node will provide self-identification and self-documentation information needed to integrate and manage the product. A network tool can read this information over the network.

4.3 Variable and Node Configuration

For discussion purposes, typically the Network is based on ANSI/EIA-709.1-A Control Network Protocol Specification, and the ANSI/EIA-709.3 Free-Topology Twisted-Pair Channel Specification (herein referenced as the "TP/FT-10 channel". This channel operates at the nominal rate of 78,125 bps on a shielded, twisted pair. The communication port conforms to the TP/FT-10 channel specification parameters.

4.3.1 Domain

The node supports 1 domain used for the application exchanges.

Table 1 Domain Index 0

Parameter	Setting
Domain ID	N/A (Zero Length Domain)
Subnet Address	1
Node Address	FB: 1, CI: 2
Length	0 bytes

4.3.2 Standard Program ID

Based on the LonMark interoperability:

9i:ii:ii:mm:ss:gu:tt:nn

Table 2 Standard Program ID Values

Field	Content	Value	Note
9i:ii:ii	Manufacture ID	90:00:79	9 indicates not certified
:mm:ss	Device Class	5A:00	No specific profile
:gu:	Device SubClass Usage	07	Transportation
:tt:	Device SubClass Channel	04	TP/FT-10
:nn:	Model	XY	See below

The last 8 bits is the model/revision number. This is a free format field as defined by LonMark. This field will represent the unit "X.Y" software version, where:

X will be 4 bits – Bit 7 thru bit 4

Y will be 4 bits – Bit 3 thru bit 0.

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4.3.3 Configuration Data Structure

The following values are used to configure the Friction equipment node.

Table 3 Configuration Data Structure

Parameter	Setting
Location	"FB"
Comm_Clock	4
Input_clock	5
Comm_type	1
Comm_pin_dir	14
Preamble_length	8
Packet_cycle	5
Beta_control	12
Xmit_interpacket	14
Rcv_interpacket	15
Node_priority	2
Channel_priority	4
Non_group_timer	5(768ms)
Nm_auth	0
Preemption_timeout	5
Collision_detect	0
Bit_sync_threshold	0
Filter	0
Hysteresis	0
Cd_tail	0
Cd_preamble	0

The following values are used to configure the Propulsion equipment node.

Table 4 Configuration Data Structure

Parameter	Setting
Channel ID	0
Location	"CI"
Comm_Clock	4
Input_clock	5
Comm_type	1
Comm_pin_dir	14
Preamble_length	8

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Parameter	Setting
Packet_cycle	5
Beta_control	12
Xmit_interpacket	14
Rcv_interpacket	15
Node_priority	3
Channel_priority	4
Non_group_timer	5(768ms)
Nm_auth	0
Preemption_timeout	5
Collision_detect	0
Bit_sync_threshold	0
Filter	0
Hysteresis	0
Cd_tail	0
Cd_preamble	0

4.3.4 Priority Slots

The LonTalk protocol offers a priority mechanism to improve the response time of critical packets. The protocol permits the user to allocate priority time slots on a channel, dedicated to priority nodes. The Propulsion and Friction Brake nodes shall each be assigned a priority slot for their critical packets. Because there is not contention for the media during the priority portion of a packet cycle, nodes configured with priority have better response time than non-priority nodes. When a priority packet is generated within a node, it travels out of the node on the priority queue, ahead of any pending non-priority packets buffered for transmission. The combination of priority and collision detection thus allows for bounded response time.

One or more of the packets sent from the Propulsion and Friction Brake nodes shall use the priority service. The Friction Brake system shall be assigned priority slot #2 and the Propulsion system shall be assigned priority slot #3. The priority designation shall be set at compile time and for the case of network variables, can be changed after installation using a network management tool. Lower priority numbers indicate higher levels of priority. For example, a priority network variable from a node with priority slot #2 will be transmitted before a priority network variable from a node with priority slot #3. (Note: Priority slot #1 is reserved for Network Management tools.)

4.3.5 Service Type

Critical messages between the Friction Brake system and the Propulsion system shall use Acknowledged Service. With Acknowledged Service, for each message sent, an individual acknowledgement is expected from the receiver. If an acknowledgement is not received from the destination, the sender times out and re-tries the transaction. The number of re-tries and the time-out are selectable. The acknowledgements are generated

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by the network CPU in the Neuron without intervention of the application. Transaction IDs are used to keep track of messages and acknowledgements so that the application does not receive duplicate messages.

After three retries, any message sent with Acknowledged Service that is not properly acknowledged shall result in a diagnostic indication to the CDT and a cessation of blending of brake, resulting in full friction brake operation, until proper communication is restored and a transition from brake to power has been commanded.

4.3.6 Network Traffic Control

Network traffic control is illustrated in Figure 1.

4.3.6.1 Minimum Send Time

This configuration property defines the minimum period of time (in seconds) between output network variable transitions. A Minimum Send Time set to zero causes no send time delay (send immediate.)

4.3.6.2 Maximum Send Time

This configuration property defines the maximum period of time (in seconds) between output network variable transitions. This property uses the SNVT_time_sec Network Variable type, and sets the maximum time between network variable updates to ensure that the latest value has been transmitted. This ensures that the network variable is updated periodically as defined by the Maximum Send Time, or earlier if a change of the network variable data satisfies the Minimum Send Time and Send on Delta criteria.

4.3.6.3 Send on Delta

This configuration property is used to determine the amount by which the value obtained by the application must change before the primary output network variable is transmitted.

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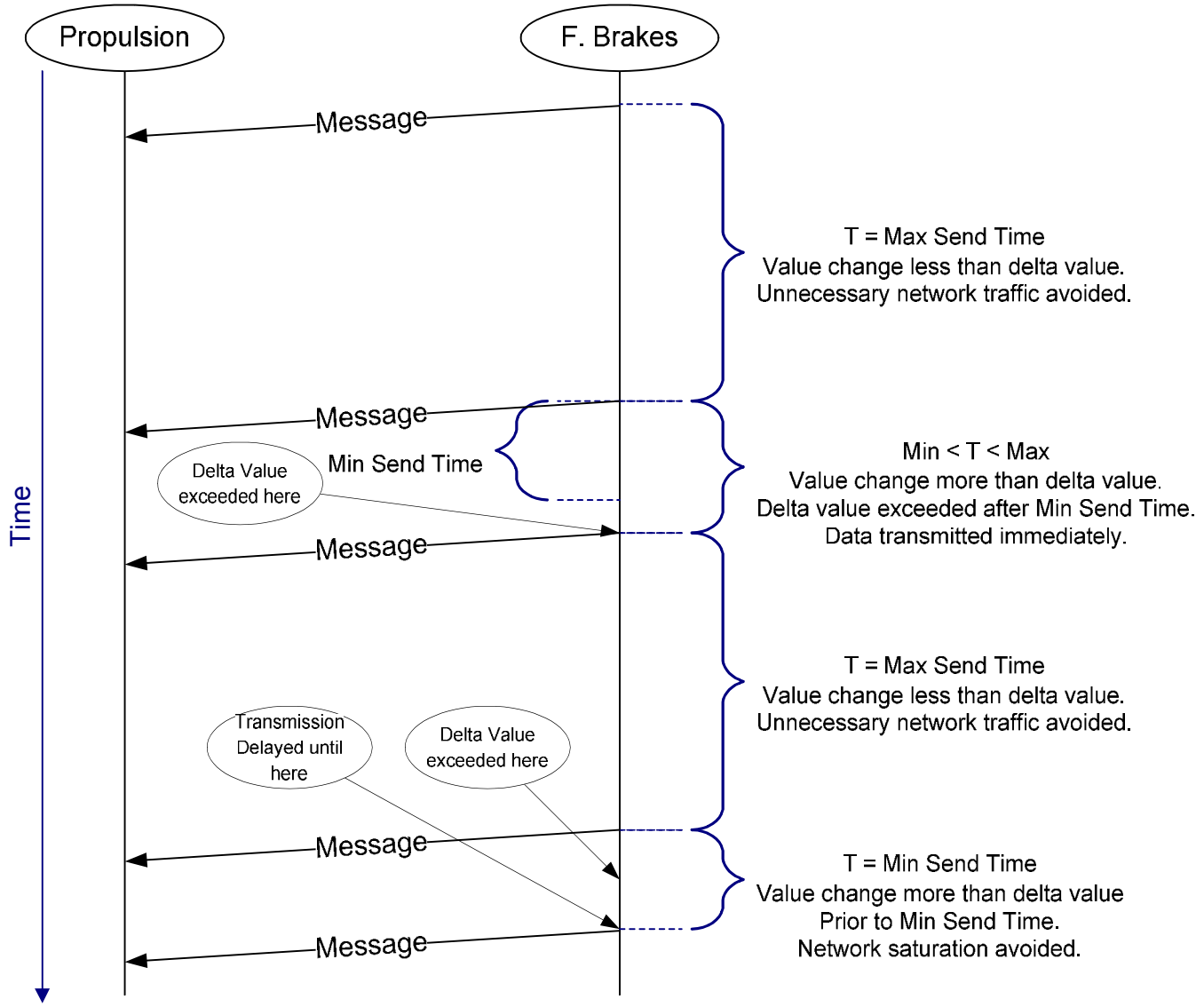

Figure 1 Network Traffic Example

4.3.7 Network Topology

The network topology shall be a bus topology as recommended in the Echelon Engineering bulletin "Design Practices for Using a LonWorks Free Topology Channel on Trains".

4.3.8 Network Wiring Interface

The network cable shall be a direct connection between the Propulsion system and the Friction Brake system.

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4.3.9 Network Cable Specifications

The network wiring shall be shielded twisted pair cable with a nominal impedance of 100 ohms as recommended in Echelons Engineering bulletin "Design Practices for Using a LonWorks Free Topology Channel on Trains".

The cable shall meet the following specifications in addition to meeting the requirements of the SEPTA Silverliner V Engineering Specification:

Impedance:	100 ohms ± 10%
Attenuation:	≤ 1 dB / 1000 ft at 100 kHz
Propagation Delay:	≤ 2 nSec / ft at 100 kHz
Mutual Capacitance:	≤ 20 pF / ft at 100 kHz
Capacitance Unbalance to Shield:	≤ 1% at 100 kHz

4.3.10 Network Termination

The bus termination shall be 100 ohm ± 1 ohm and it shall be DC isolated from the network. The network terminators will be located in the Friction Brake system. The network cable shield will be terminated at the Friction Brake system and shall not be terminated in the Propulsion systems.

4.3.11 Network Physical Layer Interface Design

The Brake to Propulsion car network shall be implemented using a LonWorks Network conforming to IEEE 1473 (Type L). The network physical layer interface design shall conform to the guidelines presented in Echelon's Engineering Bulletins: LonWorks FTT-10A Free Topology Transceiver User's Guide and Design Practices for Using a LonWorks Free Topology Channel on Trains.

4.3.12 DC Blocking Capacitor

The network physical layer shall be capable of withstanding a continuous application of a 100 VDC of either polarity. Refer to the "LonWorks FTT-10A Free Topology Transceiver User's Guide" and "Design Practices for Using a LonWorks Free Topology Channel on Trains" for details regarding AC coupling of a LonWorks network.

4.3.13 Circular Counter

Messages with data containing a circular counter shall sequentially increment the counter on each update of the network data. This update of the circular counter indicates that the data is not stale, and by monitoring the value of the circular counter, that the data is fresh. Should the circular counter not increment on three consecutive network messages, or if the absolute value of the difference between the previous and current circular counter is greater than 5 (and the heartbeat criteria is met), then this condition shall result in a diagnostic indication to the CDT and a cessation of blending of brake, resulting in full

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friction brake operation, until proper communication is restored and a transition from brake to power has been commanded.

4.3.14 Heartbeat

Messages defined as heartbeat messages shall be monitored for regular update. Should a heartbeat message not be updated before a period of twice the Maximum Send Time for that message has expired, then this condition shall result in a diagnostic indication to the CDT and a cessation of blending of brake, resulting in full friction brake operation, until proper communication is restored and a transition from brake to power has been commanded.

If all heartbeat messages are properly being received, then the circular counter shall be monitored for increasing counter values as described in section 4.3.13.

4.3.15 Installation

All nodes on this network shall be automatically self-installed upon reset. No external network manager shall be required to configure a node.

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5.0 FRICTION BRAKE OBJECT DESCRIPTION

The Friction Brake LonWorks interface is limited to:

- Load Weigh: The car weight with passenger load is provided. This signal is also used as the Heartbeat implementation from the FBCU to the MELCO Propulsion Equipment. Minimum send time is 20 msec.
- Tractive Effort Brake Command: The level of electric brake force required based on the brake pipe signal. Minimum send time is 20 msec.
- Friction Brake Control Unit Status Information:
    - The state of the FBCU wheel slip control activity. Minimum send time is 20 msec.
    - Requesting dynamic brake. Minimum send time is 20 msec.
- Tractive Effort Brake Achieved: The level of electric brake force provided by the MELCO Propulsion equipment. Minimum send time is 20 msec.
- Propulsion Status Information :
    - The state of the MELCO Propulsion wheel slip control activity. Minimum send time is 20 msec.
    - The state of the MELCO Propulsion pre-fade condition. Minimum send time is 20 msec.
    - The state of the MELCO Propulsion brake health status. This signal is also used as the Heartbeat implementation from the MELCO Propulsion Equipment to the FBCU. Minimum send time is 20 msec.

All further documentation of the LonWorks interface will be handled in the product's design documents. Data within new messages transmitted by each system must be new (current) and the system's logic must be designed such that the data transmitted is the intended data. The design documentation must clearly indicate how data is handled between the application software logic and the neuron software logic as well as any associated shared memory (RAM).

There are 6 objects (0, 1, 2, 3, 4, 5) used to realize the functionality of the LonWorks interface. The requirements for the FBCU node is shown in Figure 2 and discussed in the following sections.

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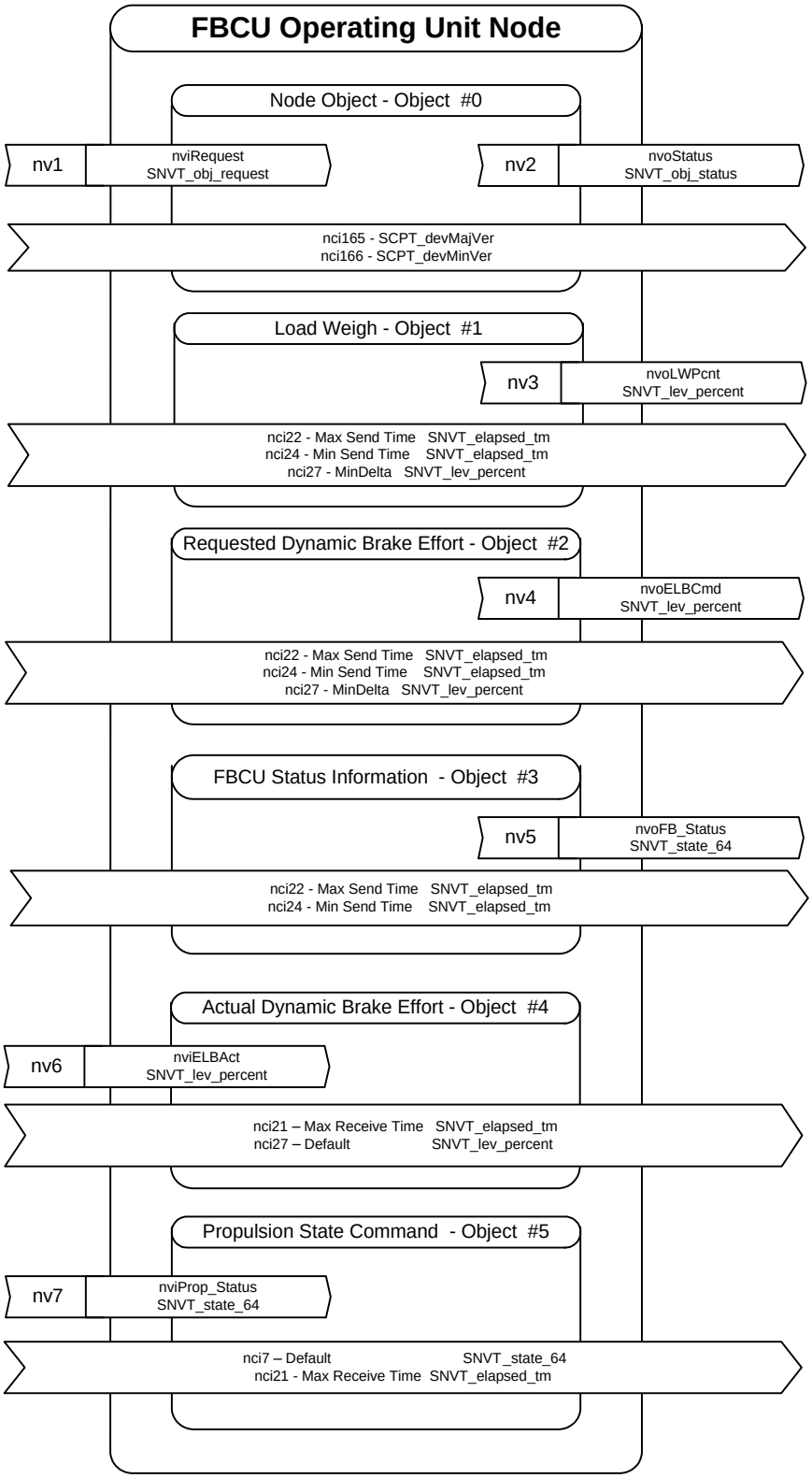

Figure 2 Friction Brake Control Unit Node

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5.1 Node Object #0

The node object provides mechanisms for requesting object modes and for reporting status of objects within the node, as well as for the node as a whole. In addition, the node object includes network variables and configuration properties relating to the complete node such as network management support.

5.1.1 nviRequest

Type: Input, SNVT_obj_request

Description: This input variable provides the mechanism to request a particular mode for the object within the node or the node as a whole. This includes requests such as: self test and update of status. This unit has no self-test capabilities.

Generally, the nviRequest input NV would have the object_id field set to relate this request to a particular object implemented on this node as per the LonMark compliance guidelines for a Node Object. This implementation deviates from that suggested behavior by assuming all input requests refer to the operating unit as a single object. The recognized values of nviRequest's object_request data member shall be limited to the values as listed in the following table.

Table 5 Object_Request Field Values

Object_request Field	Description
RQ_NORMAL	Will enable node if disabled
RQ_DISABLED	Will set node to status to disabled, and cause nvoStatus to report
RQ_UPDATE_STATUS	Will cause nvoStatus to be sent on the network
RQ_ENABLE	Will enable node if disabled
RQ_REPORT_MASK	The status bits support by the object

Refer to the SNVT Master List and Programmers Guide for details of the SNVT definition and the required data.

5.1.2 nvoStatus

Type: Output, SNVT_obj_status

Description: This output network variable reports the status for any object within the node.

The nvoStatus is output to the network to indicate the current state of the LonWorks Interface Unit. Output of this network variable shall occur only when an nviRequest RQ_UPDATE_STATUS is received.

The only recognized fields in nvoStatus that will be used are listed in the following table.

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Table 6 nvoStatus Field Values

nvoStatus Field	Description
Report_mask	Self-documenting mask describe the status bits
Invalid_id	1 means object ID not defined in the node
Invalid_request	1 means last request for unimplemented function
Disabled	1 means the object is disabled, no commands processed until enabled.

Refer to the SNVT Master List and Programmers Guide for details of the SNVT definition and the required data.

5.1.3 Configuration Properties

5.1.3.1 nciDevMajVer

Type: SCPT Index 165, SCPTdevMajVer

Description: This configuration property is used to communicate the Device Major Software Version.

5.1.3.2 nciDevMinVer

Type: SCPT Index 166, SCPTdevMinVer

Description: This configuration property is used to communicate the Devices Minor Software Version.

5.2 Object #1 – Load Weigh

5.2.1 nvoLWPcnt

Type: Output, SNVT_lev_percent, priority

Description: This output variable reports the load weigh of the car without inertia using the percent value SNVT_lev_percent. The variable is transmitted when its value has exceeded the nciMinDelta and the minimum send time has expired (as defined in the SCPT). Additionally this network variable will also be transmitted as a heartbeat output on a regular basis as dictated by the Maximum Send Time configuration nciMaxSendTime1.

Values: The minimum and maximum for the variable is normally 10 to 90% (0.10% steps). This corresponds to car weight value of 137000 lbs [62.142 metric tonnes] at 10% to 187393 lbs [85 metric tonnes] at 90%.

Error Handling: No message update after the maximum send time results in the propulsion system limiting the data field to the SEPTA TS requirement for propulsion and brake mode default values, as specified in sections 1.28.7 and 11.4.2. Data values out of the expected range specified will

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be considered invalid and handled independently by each subsystem as required by the SEPTA Technical Specification, by limiting the valid value to the expected values between AW0 and AW3.

5.2.2 **Configuration Properties**

5.2.2.1 nciMaxSendTime1

Type: SCPT Index 22, SNVT_elapsed_tm

Description: Indicates the maximum period of time that expires before the object automatically updates its current output variables. The network management tool can change this property up to the maximum defined as 65000 seconds, with values greater than 65000 seconds limited to 65000 seconds. Setting this SCPT to 0 disables this controlled variable from being sent.

Preset Value: 200 milliseconds

5.2.2.2 nciMinSendTime1

Type: SCPT Index 24, SNVT_elapsed_tm

Description: Indicates the minimum period between output network variable transitions. In other words, the appropriate network variable output will not be updated faster than specified by nciMinSendT1 when a change occurs. Note: Status changes that occur faster than the minimum send time could be missed. This SCPT can be set to 0 which allows the unit to send the controlled variable on updates only. Updates are based then on the MinDelta setting.

Preset Value: 20 milliseconds

5.2.2.3 nciMinDelta1

Type: SCPT Index 27, SNVT_lev_percent

Description: Indicates the minimum percent change required to update the output network variable.

Preset Value: SNVT value change is 100 counts (0.5%)

5.3 <u>Object #2 – Requested Dynamic Brake Effort</u>

5.3.1 **nvoELBCmd**

Type: Output, SNVT_lev_percent, priority

Description: This output variable reports the requested dynamic brake effort force (Electric Brake) using the percent value SNVT_lev_percent. The variable is transmitted when its value has exceeded the nciMinDelta and

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after the min send time has expired (as defined in the SCPT). Additionally this network variable will also be transmitted on a regular basis as dictated by the Maximum Send Time configuration nciMaxSendTime2.

Values: The minimum and maximum for the variable is normally 10 to 90% (0.10% steps). This corresponds to forces of 0 lbs [0 kN] to 26977 lbs [120 kN]. The default out of range value is 0% when the device is known to be defective and any value outside the normal range specified which is considered to be invalid and result in 0 lbs [0 kN]. The propulsion system must ensure any change in the achieved effort satisfies the jerk limit requirement as stated in SEPTA TS 1.28.8.

Error Handling: Transmission errors or no message update after the maximum send time has expired, results in the propulsion system jerk limiting the current effort to a coast (no brake effort) condition. Data values outside of expected range are handled by limiting the valid value to the expected endpoint values prior to transmitting.

5.3.2 Configuration Properties

5.3.2.1 nciMaxSendTime2

Type: SCPT Index 22, SNVT_elapsed_tm

Description: Indicates the maximum period of time that expires before the object automatically updates its current output variables. The network management tool can change this property up to the maximum defined as 65000 seconds, with values greater than 65000 seconds limited to 65000 seconds. Setting this SCPT to 0 disables this controlled variable from being sent.

Preset Value: 200 milliseconds

5.3.2.2 nciMinSendTime2

Type: SCPT Index 24, SNVT_elapsed_tm

Description: Indicates the minimum period between output network variable transitions. In other words, the appropriate network variable output will not be updated faster than specified by nciMinSendT2 when a change occurs. Note: Status changes that occur faster than the minimum send time could be missed. This SCPT can be set to 0 which allows the unit to send the controlled variable on updates only. Updates are based then on the MinDelta setting.

Preset Value: 20 milliseconds

5.3.2.3 nciMinDelta2

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Type: SCPT Index 27, SNVT_lev_percent

Description: Indicates the minimum percent change required to update the output network variable.

Preset Value: SNVT value change is 60 counts (0.3%)

5.4 Object #3 - FBCU Status Information

This object is to provide the FBCU status information to the MELCO Propulsion Equipment.

5.4.1 **nvoFB_Status**

Type: Output, SNVT_state_64, priority

Description: This output variable reports various status information about the FBCU Equipment using the SNVT_state_64. This network variable will be transmitted as a heartbeat output on a regular basis as dictated by the Maximum Send Time configuration nciMaxSendTime3. If a change occurs then it will be sent at the Minimum Send Time. Information is provided about the status below.

Table 7 nvoFB_Status Bit Definition

Bit	Definition	Comment
0 MSB	1 = Wheel Slip Activity (Detected)	
1	1 = Requesting Dynamic Brake Effort 0 = Not Requesting Dynamic Brake Effort	The signal state must correspond with the demand SNVT. Whenever demand force is greater than 0 lbs [0 kN], this bit must be ‘1’ to indicate dynamic effort it requested. The propulsion electric brake C/I must confirm this condition to provide the braking effort requested, otherwise 0 lbs [0 kN] effort is provided by propulsion brake (friction only mode). Diagnostic tolerance for confirmation of the information to be defined by each subsystem and finalized during car testing, typical expected timing not to exceed 1000 msec.

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Bit	Definition	Comment
2	1 = Brake Effort mode 0 = Coast mode	The signal corresponds with the brake mode condition which indicates to the propulsion electric brake C/I that the braking is active. Otherwise the mode is coast. This bit status allows the C/I to change from coasting mode to braking mode in preparation for responding to the demand request during blending and slide control.
3 – 55	Not Used	
56-63	0-255, wrapping	Circular counter to verify sequential messaging

5.4.2 Configuration Properties

5.4.2.1 nciMaxSendTime3

Type: SCPT Index 22, SNVT_elapsed_tm

Description: Indicates the maximum period of time that expires before the object automatically updates its current output variables. The network management tool can change this property up to the maximum defined as 65000 seconds, with values greater than 65000 seconds limited to 65000 seconds. Setting this SCPT to 0 disables this controlled variable from being sent.

Preset Value: 200 milliseconds

5.4.2.2 nciMinSendTime3

Type: SCPT Index 24, SNVT_elapsed_tm

Description: Indicates the minimum period between output network variable transitions. In other words, the appropriate network variable output will not be updated faster than specified by nciMinSendT3 when a change occurs. Note: Status changes that occur faster than the minimum send time could be missed. This SCPT can be set to 0 which allows the unit to send the controlled variable on updates only.

Preset Value: 20 milliseconds

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5.5 Object #4 – Actual Dynamic Brake Effort

5.5.1 **nviELBAct**

Type: Input, SNVT_lev_percent, priority

Description: This input variable reports the actual dynamic brake effort (Electric Brake) using the percent value SNVT_lev_percent. The variable is transmitted by the propulsion equipment to the FBCU when its value has exceeded the nciMinDelta and after the min send time has expired (as defined in the SCPT). Additionally this network variable will also be transmitted on a regular basis as dictated by the Maximum Send Time configuration nciMaxSendTime4.

Values: The minimum and maximum for the variable is normally 10 to 90% (0.10% steps). This corresponds to forces of 0 lbs [0 kN] to 26977 lbs [120 kN]. The default out of range value is 0% when the device is known to be defective and any value outside the normal range specified which is considered to be invalid and result in 0 lbs [0 kN]. The propulsion system must ensure any change in the achieved effort satisfies the jerk limit requirement as stated in SEPTA TS 1.28.8.

Error Handling: Transmission errors or no message update after the maximum receive time has expired, results in the friction brake system commanding the current effort to a coast (no brake effort) condition and providing friction only brake. Data values sent out of expected range shall be interpreted by the FBEU as 0 kN, and the corresponding command shall be set to 0 kN for full friction braking for the remainder of the brake condition.

5.5.2 **Configuration Properties**

5.5.2.1 nciDefault

Type: SCPT Index 27, SNVT_lev_percent

Description: This configuration property determines the percent adopted when no updates are received by nviELBAct within the maximum receive time, at power-on or reset.

Preset Value: The default will be set to zero.

5.5.2.2 nciMaxReceiveT4

Type: SCPT Index 21, SNVT_elapsed_tm

Description: This configuration property is used to control the maximum time that elapses after the last update to nviELBAct before adopting the Default value. The network management tool can change this property to the maximum defined for nciMaxReceiveT4. The network management tool can change this property up to the

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maximum defined as 65000 seconds, with values greater than 65000 seconds limited to 65000 seconds.

Preset Value: 500 millisecond

5.6 Object #5 - Propulsion Status Information

This object is to provide status information about the Propulsion Equipment to the FBCU. This information will also be used as the Heartbeat to the FBCU.

5.6.1 nviProp_Status

Type: Input, SNVT_state_64, priority

Description: This input variable reports various status information about the MELCO Propulsion Equipment using the SNVT_state_64. This network variable will be transmitted as a heartbeat output on a regular basis as dictated by the Maximum Send Time configuration nciMaxSendTime5 for the Propulsion Equipment. Information is provided about the status below in the reference information section.

Table 8 nviProp_Status Bit Definition

Bit	Definition	Comment
0 MSB	1 = Propulsion system Wheel Slip Activity (Detected) Truck A End	
1	1 = Propulsion system Wheel Slip Activity (Detected) Truck F End	
2	1 = Propulsion Fade Out Active	
3	0 = Propulsion Braking is in Failed mode	No Electric Brake Effort
4	1 = Providing Dynamic Brake Effort 0 = None	The signal state must correspond with the actual achieved SNVT value. The friction brake system must confirm that this bit and the actual effort information are consistent for blending to take place, otherwise effort is considered 0 lbs [0 kN]. Diagnostic tolerance for confirmation of the information to be defined by each subsystem and finalized during car testing, typical expected timing is not to exceed 1000 msec.

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Bit	Definition	Comment
5 – 55	Not Used	
56-63	0-255, wrapping	Circular counter to verify sequential messaging

5.6.2 Configuration Properties

5.6.2.1 nciDefault

Type: SCPT Index 7, SNVT_state_64

Description: This configuration property determines the state adopted when no updates are received by nviPropStatus within the maximum receive time, at power-on or reset.

Preset Value: The default will be all bits set to zero.

5.6.2.2 nciMaxReceiveT5

Type: SCPT Index 21, SNVT_elapsed_tm

Description: This configuration property is used to control the maximum time that elapses after the last update to nviPropStatus before adopting the Default state. The network management tool can change this property to the maximum defined for nciMaxReceiveT5. The network management tool can change this property up to the maximum defined as 65000 seconds, with values greater than 65000 seconds limited to 65000 seconds.

Preset Value: 500 millisecond

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6.0 PROPULSION NODE OBJECT DESCRIPTION

The Propulsion Brake LonWorks interface is limited to:

- Load Weigh: The car weight with passenger load is provided. This signal is also used as the Heartbeat implementation from the FBCU to the MELCO Propulsion Equipment. Minimum send time is 20 msec.
- Tractive Effort Brake Command: The level of electric brake force required based on the brake pipe signal. Minimum send time is 20 msec.
- Friction Brake Control Unit Status Information:
    - The state of the FBCU wheel slip control activity. Minimum send time is 20 msec.
    - Requesting dynamic brake. Minimum send time is 20 msec.
- Tractive Effort Brake Achieved: The level of electric brake force provided by the MELCO Propulsion equipment. Minimum send time is 20 msec.
- Propulsion Status Information :
    - The state of the MELCO Propulsion wheel slip control activity. Minimum send time is 20 msec.
    - The state of the MELCO Propulsion pre-fade condition. Minimum send time is 20 msec.
    - The state of the MELCO Propulsion brake health status. This signal is also used as the Heartbeat implementation from the MELCO Propulsion Equipment to the FBCU. Minimum send time is 20 msec.

All further documentation of the LonWorks interface will be handled in the product's design documents. Data within new messages transmitted by each system must be new (current) and the system's logic must be designed such that the data transmitted is the intended data. The design documentation must clearly indicate how data is handled between the application software logic and the neuron software logic as well as any associated shared memory (RAM).

There are 6 objects (0, 1, 2, 3, 4, 5) used to realize the functionality of the LonWorks interface. The requirements for MELCO Propulsion Equipment node is shown in Figure 3 and discussed in the following sections.

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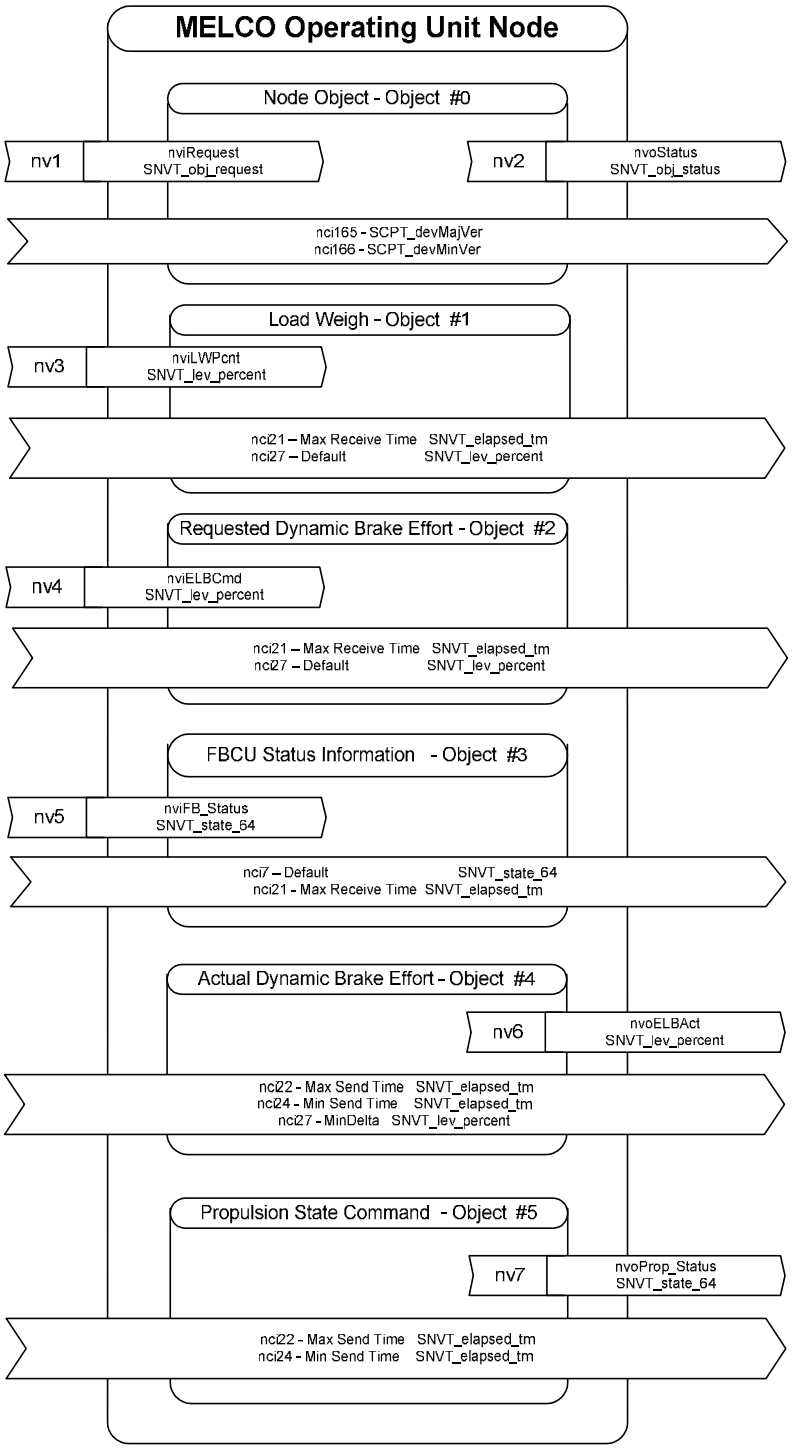

Figure 3 Propulsion Node

6.1 Node Object #0

The node object provides mechanisms for requesting object modes and for reporting status of objects within the node, as well as for the node as a whole. In addition, the node object includes network variables and configuration properties relating to the complete node such as network management support.

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6.1.1 nviRequest

Type: Input, SNVT_obj_request

Description: This input variable provides the mechanism to request a particular mode for the object within the node or the node as a whole. This includes requests such as: self test and update of status. This unit has no self-test capabilities.

Generally, the nviRequest input NV would have the object_id field set to relate this request to a particular object implemented on this node as per the LonMark compliance guidelines for a Node Object. This implementation deviates from that suggested behavior by assuming all input requests refer to the operating unit as a single object. The recognized values of nviRequest's object_request data member shall be limited to the values as listed in the following table.

Table 9 Object_Request Field Values

Object_request Field	Description
RQ_NORMAL	Will enable node if disabled
RQ_DISABLED	Will set node to status to disabled, and cause nvoStatus to report
RQ_UPDATE_STATUS	Will cause nvoStatus to be sent on the network
RQ_ENABLE	Will enable node if disabled
RQ_REPORT_MASK	The status bits support by the object

Refer to the SNVT Master List and Programmers Guide for details of the SNVT definition and the required data.

6.1.2 nvoStatus

Type: Output, SNVT_obj_status

Description: Output network variable reports the status for any object within the node.

The nvoStatus is output to the network to indicate the current state of the equipment. Output of this network variable shall occur only when an nviRequest RQ_UPDATE_STATUS is received.

The only recognized fields in nvoStatus that will be used are listed in the following table.

Table 10 nvoStatus Field Values

nvoStatus Field	Description
Report_mask	Self-documenting mask describe the status bits
Invalid_id	1 means object ID not defined in the node
Invalid_request	1 means last request for unimplemented function
Disabled	1 means the object is disabled, no commands processed until enabled.

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Refer to the SNVT Master List and Programmers Guide for details of the SNVT definition and the required data.

6.1.3 **Configuration Properties**

6.1.3.1 nciDevMajVer

Type: SCPT Index 165, SCPTdevMajVer

Description: This configuration property is used to communicate the Device Major Software Version.

6.1.3.2 nciDevMinVer

Type: SCPT Index 166, SCPTdevMinVer

Description: This configuration property is used to communicate the Devices Minor Software Version.

6.2 Object #1 – Load Weigh

6.2.1 **nviLWPcnt**

Type: Input, SNVT_lev_percent, priority

Description: This input variable reports the load weigh of the car using the percent value SNVT_lev_percent. The variable is transmitted by the FBCU when its value has exceeded the nciMinDelta and after the minimum send time has expired (as defined in the SCPT). Additionally this network variable will also be transmitted as a heartbeat output on a regular basis as dictated by the Maximum Send Time configuration nciMaxSendTime1 for the FBCU.

Values: The minimum and maximum for the variable is normally 10 to 90% (0.10% steps). This corresponds to car weight value of 137000 lbs [62.142 metric tonnes] at 10% to 187393 lbs [85 metric tonnes] at 90%.

Error Handling: No message update after the maximum receive time results in the propulsion system limiting the data field to the SEPTA TS requirement for propulsion and brake mode default values. Data values out of the expected range specified will be considered invalid and handled independently by each subsystem as required by the SEPTA Technical Specification.by limiting the valid value to the expected values between AW0 and AW3.

6.2.2 **Configuration Properties**

6.2.2.1 nciDefault

Type: SCPT Index 27, SNVT_lev_percent

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Description: This configuration property determines the percent adopted when no updates are received by nviLWPcnt within the maximum receive time, at power-on or reset.

Preset Value: The default will be set to zero.

6.2.2.2 nciMaxReceiveT1

Type: SCPT Index 21, SNVT_elapsed_tm

Description: This configuration property is used to control the maximum time that elapses after the last update to nviLWPcnt before adopting the Default value. The network management tool can change this property to the maximum defined for nciMaxReceiveT1. The network management tool can change this property up to the maximum defined as 65000 seconds, with values greater than 65000 seconds limited to 65000 seconds.

Preset Value: 500 millisecond

6.3 Object #2 – Requested Dynamic Brake Effort

6.3.1 **nviELBCmd**

Type: Input, SNVT_lev_percent, priority

Description: This input variable reports the requested dynamic brake effort force (Electric Brake) using the percent value SNVT_lev_percent. The variable is transmitted by the FBCU when its value has exceeded the nciMinDelta and after the min send time has expired (as defined in the SCPT). Additionally this network variable will also be transmitted on a regular basis as dictated by the Maximum Send Time configuration nciMaxSendTime2 for the FBCU.

Values: The minimum and maximum for the variable is normally 10 to 90% (0.10% steps). This corresponds to forces of 0 lbs [0 kN] to 26977 lbs [120 kN]. The default out of range value is 0% when the device is known to be defective and any value outside the normal range specified which is considered to be invalid and result in 0 lbs [0 kN]. The propulsion system must ensure any change in the achieved effort satisfies the jerk limit requirement as stated in SEPTA TS 1.28.8.

Error Handling: Transmission errors or no message update after the maximum receive time has expired, results in the propulsion system jerk limiting the current effort to a coast (no brake effort, 0 lbs [0 kN]) condition. Data values received outside of expected range are shall be interpreted by the propulsion system as 0 lbs [0 kN].

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6.3.2 Configuration Properties

6.3.2.1 nciDefault

Type: SCPT Index 27, SNVT_lev_percent

Description: This configuration property determines the percent adopted when no updates are received by nviELBCmd within the maximum receive time, at power-on or reset.

Preset Value: The default will be set to zero.

6.3.2.2 nciMaxReceiveT2

Type: SCPT Index 21, SNVT_elapsed_tm

Description: This configuration property is used to control the maximum time that elapses after the last update to nviELBCmd before adopting the Default value. The network management tool can change this property to the maximum defined for nciMaxReceiveT2, which is less than the SNVT used. The network management tool can change this property up to the maximum defined as 65000 seconds, with values greater than 65000 seconds limited to 65000 seconds.

Preset Value: 500 millisecond

6.4 Object #3 - FBCU Status Information

This object is to provide the FBCU status information to the MELCO Propulsion Equipment.

6.4.1 nviFB_Status

Type: Input, SNVT_state_64, priority

Description: This input variable reports various status information about the FBCU Equipment using the SNVT_state_64. This network variable will be transmitted as a heartbeat output on a regular basis as dictated by the Maximum Send Time configuration nciMaxSendTime1. If a change occurs then it will be sent at the Minimum Send Time. Information is provided about the status below in the reference information section.

Table 11 nvoFB_Status Bit Definition

Bit	Definition	Comment
0 MSB	1 = Wheel Slip Activity (Detected)	
1	1 = Requesting Dynamic Brake Effort 0 = Not Requesting Dynamic Brake Effort	The signal state must correspond with the demand SNVT. Whenever demand force is greater than 0 lbs [0

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Bit	Definition	Comment
		kN], this bit must be '1' to indicate dynamic effort it requested. The propulsion electric brake C/I must confirm this condition to provide the braking effort requested, otherwise 0 lbs [0 kN] effort is provided by propulsion brake (friction only mode). Diagnostic tolerance for confirmation of the information to be defined by each subsystem and finalized during car testing, typical expected timing not to exceed 1000 msec.
2	1 = Brake Effort mode 0 = Coast mode	The signal corresponds with the brake mode condition which indicates to the propulsion electric brake C/I that the braking is active. Otherwise the mode is coast. This bit status allows the C/I to change from coasting mode to braking mode in preparation for responding to the demand request during blending and slide control.
3-55	Spare	
56-63	0-255, wrapping	Circular counter to verify sequential messaging

6.4.2 Configuration Properties

6.4.2.1 nciDefault

Type: SCPT Index 7, SNVT_state_64

Description: This configuration property determines the state adopted when no updates are received by nviFB_Status within the maximum receive time, at power-on or reset.

Preset Value: The default will be all bits set to zero.

6.4.2.2 nciMaxReceiveT3

Type: SCPT Index 21, SNVT_elapsed_tm

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Description: This configuration property is used to control the maximum time that elapses after the last update to nviFB_Status before adopting the Default values. The network management tool can change this property to the maximum defined for nciMaxReceiveT3, which is less than the SNVT used. The network management tool can change this property up to the maximum defined as 65000 seconds, with values greater than 65000 seconds limited to 65000 seconds.

Preset Value: 500 millisecond

6.5 Object #4 – Actual Dynamic Brake Effort

6.5.1 **nvoELBAct**

Type: Output, SNVT_lev_percent, priority

Description: This output variable reports the actual dynamic brake effort (Electric Brake) using the percent value SNVT_lev_percent. The variable is transmitted to the FBCU when its value has exceeded the nciMinDelta and after the min send time has expired (as defined in the SCPT). Additionally this network variable will also be transmitted on a regular basis as dictated by the Maximum Send Time configuration nciMaxSendTime4.

Values: The minimum and maximum for the variable is normally 10 to 90% (0.10% steps). This corresponds to forces of 0 lbs [0 kN] to 26977 lbs [120 kN]. The default out of range value is 0% when the device is known to be defective and any value outside the normal range specified which is considered to be invalid and result in 0 lbs [0 kN]. The actual effort includes jerk limiting, as the propulsion system is expected to perform the jerk limiting task in relationship to both the SEPTA TS jerk limit requirements of -1.00 to -3.00 mphpsps and the response characteristics of the pneumatic system.

Error Handling: Transmission errors or no message update after the maximum receive time has expired, results in the friction brake system commanding the current effort to a coast (no brake effort) condition and providing friction only brake. Data values sent out of expected range shall be interpreted by the FBEU as 0 lbs [0 kN].

6.5.2 **Configuration Properties**

6.5.2.1 nciMaxSendTime4

Type: SCPT Index 22, SNVT_elapsed_tm

Description: Indicates the maximum period of time that expires before the object automatically updates its output variables. The network management tool can change this property up to the maximum

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defined as 65000 seconds, with values greater than 65000 seconds limited to 65000 seconds. Setting this SCPT to 0 disables this controlled variable from being sent.

Preset Value: 200 milliseconds

6.5.2.2 nciMinSendTime4

Type: SCPT Index 24, SNVT_elapsed_tm

Description: Indicates the minimum period between output network variable transitions. In other words, the appropriate network variable output will not be updated faster than specified by nciMinSendT4 when a change occurs. Note: Status changes that occur faster than the minimum send time could be missed. This SCPT can be set to 0 which allows the unit to send the controlled variable on updates only. Updates are based then on the MinDelta setting.

Preset Value: 20 milliseconds

6.5.2.3 nciMinDelta3

Type: SCPT Index 27, SNVT_lev_percent

Description: Indicates the minimum percent change required to update the output network variable. The percent change is based on the maximum range (full scale) for the valid range in kN.

Preset Value: SNVT value change is 60 (0.3%)

6.6 Object #5 - Propulsion Status Information

This object is to provide status information about the Propulsion Equipment to the FBCU. This information will also be used as the Heartbeat to the FBCU.

6.6.1 **nvoProp_Status**

Type: Output, SNVT_state_64, priority

Description: This output variable reports various status information about the MELCO Propulsion Equipment using the SNVT_state_64. This network variable will be transmitted as a heartbeat output on a regular basis as dictated by the Maximum Send Time configuration nciMaxSendTime5. If a change occurs then it will be sent at the Minimum Send Time. Information is provided for the status below.

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Table 12 nviProp_Status Bit Definition

Bit	Definition	Comment
0 MSB	1 = Wheel Slip Activity (Detected) Truck A End	
1	1 = Wheel Slip Activity (Detected) Truck F End	
2	1 = Fade Out Active	
3	0 = Propulsion Braking is in Failed mode	No Electric Brake Effort
4	1 = Providing Dynamic Brake Effort 0 = None	The signal state must correspond with the actual achieved SNVT value. The friction brake system must confirm that this bit and the actual effort information are consistent for blending to take place, otherwise effort is considered 0 lbs [0 kN]. Diagnostic tolerance for confirmation of the information to be defined by each subsystem and finalized during car testing, typical expected timing not to exceed 1000 msec.
5 – 55	Not Used	
56-63	0-255, wrapping	Circular counter to verify sequential messaging

6.6.2 Configuration Properties

6.6.2.1 nciMaxSendTime5

Type: SCPT Index 22, SNVT_elapsed_tm

Description: Indicates the maximum period of time that expires before the object automatically updates its output variables. The network management tool can change this property up to the maximum defined as 65000 seconds, with values greater than 65000 seconds limited to 65000 seconds. Setting this SCPT to 0 disables this controlled variable from being sent.

Preset Value: 200 mseconds

6.6.2.2 nciMinSendTime5

Type: SCPT Index 24, SNVT_elapsed_tm

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Description: Indicates the minimum period between output network variable transitions. In other words, the appropriate network variable output will not be updated faster than specified by nciMinSendT5 when a change occurs. Note: Status changes that occur faster than the minimum send time could be missed. This SCPT can be set to 0 which allows the unit to send the controlled variable on updates only.

Preset Value: 20 milliseconds

7.0 NETWORK VARIABLES

Table 13 FBCU Network Variables

Name	SNVT index	Type	Service Type[1]	Index Address	Selector Value[2]	Addressing	Period Send Max	Period Send Min
nviRequest	92	SNVT_obj_request	ACKD	0	0x1600	Subnet/Node	-	-
nvoStatus	93	SNVT_obj_status	ACKD	1	0x1601	Subnet/Node	-	-
nvoLWPcnt	81	SNVT_lev_percent	UNACKD	2	0x1602	Subnet/Node	200 ms	20 ms
nvoELBCmd	81	SNVT_lev_percent	ACKD	3	0x1603	Subnet/Node	200 ms	20 ms
nvoFB_Status	165	SNVT_state_64	UNACKD	4	0x1604	Subnet/Node	200 ms	20 ms
nviELBAct	81	SNVT_lev_percent	ACKD	5	0x1605	Subnet/Node	200 ms	20 ms
nviProp_Status	165	SNVT_state_64	UNACKD	6	0x1606	Subnet/Node	200 ms	20 ms
nciMaxSendTime1	87	SNVT_elapsed_tm	UNACKD	7	0x1607	Subnet/Node	-	-
nciMinSendTime1	87	SNVT_elapsed_tm	UNACKD	8	0x1608	Subnet/Node	-	-
nciMaxSendTime2	87	SNVT_elapsed_tm	UNACKD	9	0x1609	Subnet/Node	-	-
nciMinSendTime2	87	SNVT_elapsed_tm	UNACKD	10	0x160A	Subnet/Node	-	-
nciMaxSendTime3	87	SNVT_elapsed_tm	UNACKD	11	0x160B	Subnet/Node	-	-
nciMinSendTime3	87	SNVT_elapsed_tm	UNACKD	12	0x160C	Subnet/Node	-	-
nciMaxReceiveT4	87	SNVT_elapsed_tm	UNACKD	13	0x160D	Subnet/Node		
nciMaxReceiveT5	87	SNVT_elapsed_tm	UNACKD	14	0x160E	Subnet/Node		
nciMinDelta1	81	SNVT_lev_percent	UNACKD	15	0x160F	Subnet/Node	-	-
nciMinDelta2	81	SNVT_lev_percent	UNACKD	16	0x1610	Subnet/Node	-	-
nciDefault4	165	SNVT_state_64	UNACKD	17	0x1611	Subnet/Node	-	-
nciDefault5	165	SNVT_state_64	UNACKD	18	0x1612	Subnet/Node		
nciDevMajVer		Unsigned Short	UNACKD	19	0x1613	Subnet/Node		
nciDevMinVer		Unsigned Short	UNACKD	20	0x1614	Subnet/Node		

Table 14 Propulsion Network Variables

Name	SNVT index	Type	Service Type[1]	Index Address	Selector Value[2]	Addressing	Period Max	Period Min
nviRequest	92	SNVT_obj_request	ACKD	3	0x1620	Subnet/Node	-	-
nvoStatus	93	SNVT_obj_status	ACKD	5	0x1621	Subnet/Node	-	-
nviLWPcnt	81	SNVT_lev_percent	UNACKD	10	0x1602	Subnet/Node	200 ms	20 ms
nviELBCmd	81	SNVT_lev_percent	ACKD	13	0x1603	Subnet/Node	200 ms	20 ms
nviFB_Status	165	SNVT_state_64	UNACKD	16	0x1604	Subnet/Node	200 ms	20 ms
nvoELBAct	81	SNVT_lev_percent	ACKD	20	0x1605	Subnet/Node	200 ms	20 ms
nvoProp_Status	165	SNVT_state_64	UNACKD	23	0x1606	Subnet/Node	200 ms	20 ms
nciMaxSendTime4	87	SNVT_elapsed_tm	UNACKD	17	0x1627	Subnet/Node	-	-
nciMinSendTime4	87	SNVT_elapsed_tm	UNACKD	18	0x1628	Subnet/Node	-	-
nciMaxSendTime5	87	SNVT_elapsed_tm	UNACKD	21	0x1629	Subnet/Node	-	-
nciMinSendTime5	87	SNVT_elapsed_tm	UNACKD	22	0x162A	Subnet/Node	-	-

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Name	SNVT index	Type	Service Type[1]	Index Address	Selector Value[2]	Addressing	Period Max	Period Min
nciMinDelta3	81	SNVT_lev_percent	UNACKD	19	0x162B	Subnet/Node	-	-
nciMaxReceiveT1	87	SNVT_elapsed_tm	UNACKD	9	0x162C	Subnet/Node	-	-
nciMaxReceiveT2	87	SNVT_elapsed_tm	UNACKD	11	0x162D	Subnet/Node		
nciMaxReceiveT3	87	SNVT_elapsed_tm	UNACKD	14	0x162E	Subnet/Node		
nciDefault1	81	SNVT_lev_percent	UNACKD	8	0x162F	Subnet/Node	-	-
nciDefault2	81	SNVT_lev_percent	UNACKD	12	0x1630	Subnet/Node	-	-
nciDefault3	165	SNVT_state_64	UNACKD	15	0x1631	Subnet/Node	-	-
nciDevMajVer		Unsigned Short	UNACKD	1	0x1632	Subnet/Node		
nciDevMinVer		Unsigned Short	UNACKD	2	0x1633	Subnet/Node		

Notes:

1. The network configuration tool for the network variables assigns the service types. Values listed are to be first set during manufacturing of the node. Any changes to this will be done at car installation. This implementation will support all four classes of service: ACKD, UNACKD, UNACKD_RPT, and REQUEST.

2. These are the expected values. Values listed are to be first set during manufacturing of the node.

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